RAFFLES INSTITUTION

2026 YEAR 6 PRELIMINARY EXAMINATION

Higher 2

CANDIDATE NAME

CLASS | 2 | 6 | S | 0 | 2 | B |

PHYSICS **9478/02**

Paper 2 Structured Questions **16 September 2026**

2 hours

You must answer on the question paper.

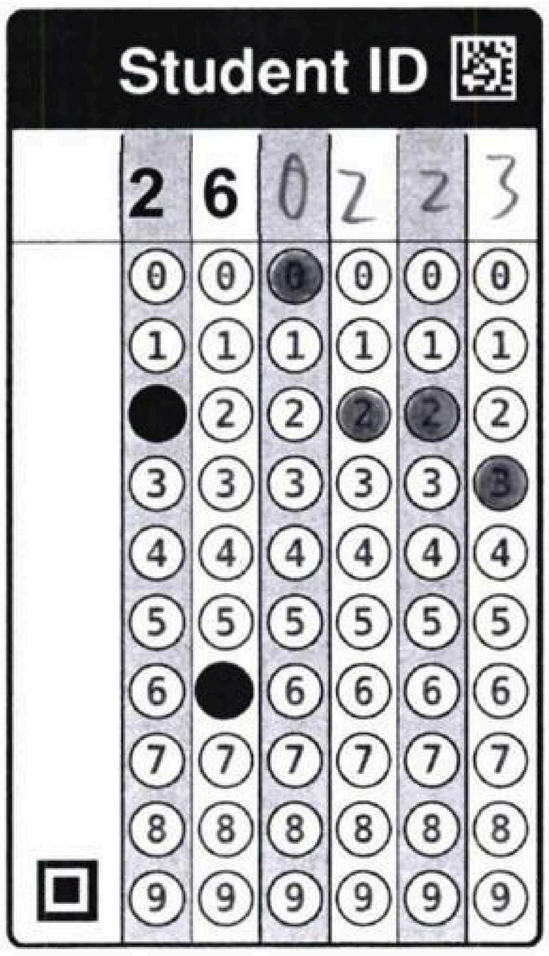
No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use a 2B pencil for any diagrams or graphs.
- Write your name and class in the spaces at the top of the page.
- Shade the lozenges in the Student ID box below according to your RI Exam Index No.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- This section carries a total of 75 marks.
- The number of marks for each question or part question is shown in brackets [ ].



For Examiner's Use	
1	5 / 9
2	1 / 6
3	0 / 10
4	1 / 8
5	0 / 5
6	0 / 4
7	2 / 11
8	8 / 22
Deduction	0
Total	17 / 75

This document has **22** pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi\times10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left.8.99\times10^{9}\text{ m F}^{-1}\right)$
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut+\frac{1}{2}at^2$
	v^2	=	u^2+2as
work done on / by a gas	W	=	$p\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T/K	=	$T/°\text{C}+273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2\rangle$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$

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displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 \mathrm{I}}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N \mathrm{I}}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 n \mathrm{I}$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2} n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
radioactive decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** A ball of mass 0.80 kg is released from rest in air from a tall building. The variation of the velocity of ball v with time t is shown in Fig. 1.1.

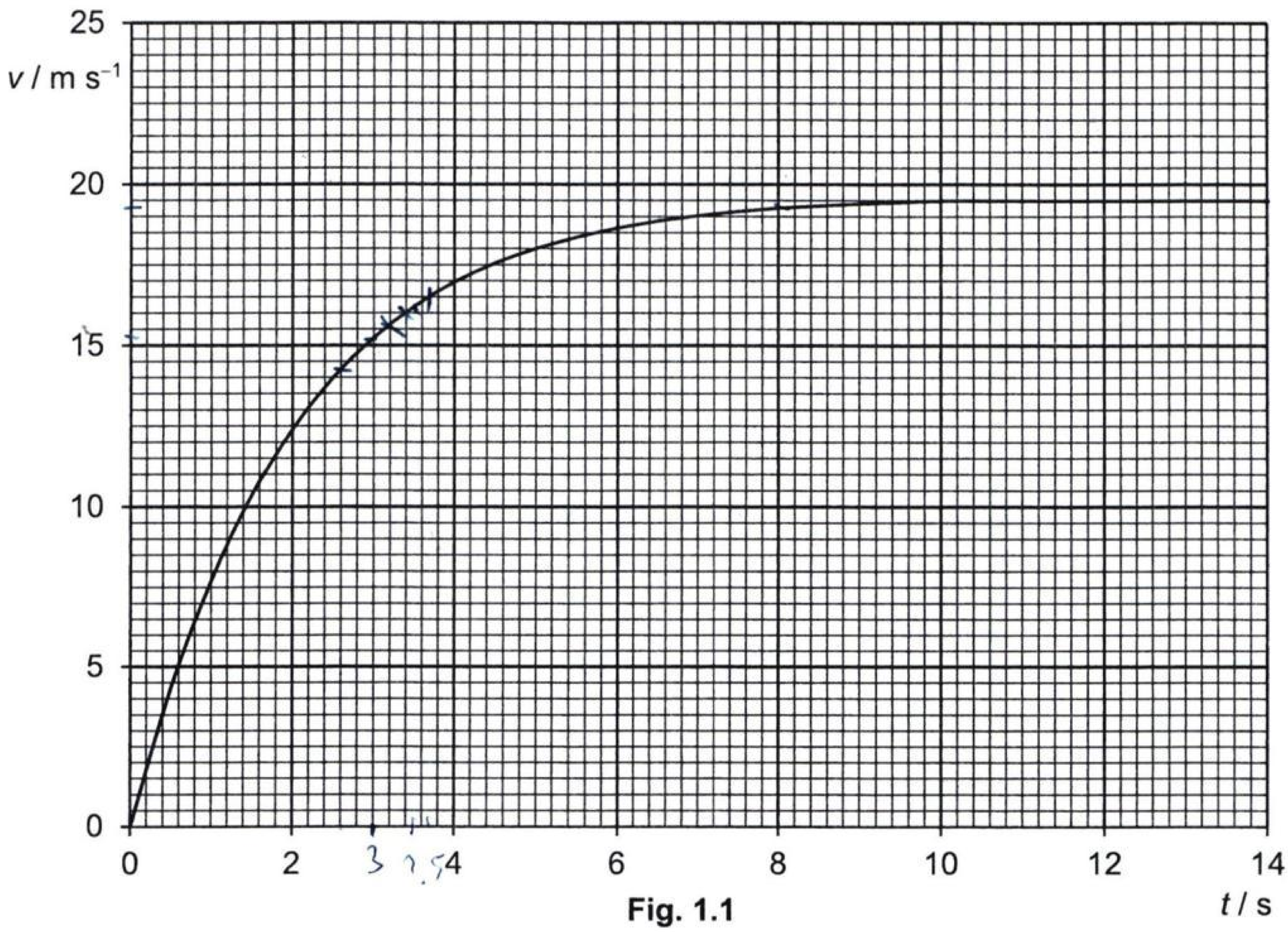


Fig. 1.1

(i) With reference to Fig. 1.1, explain how the drag force F_D acting on the ball changes as the velocity of ball increases from t = 0 to t = 8.0 s.

1 / 2

The drag force is directly proportional to the velocity, ~~in the~~ opposing the direction of velocity. By newton's second law, F=ma and the force in opposite direction has an effect on the velocity such that the velocity ~~de~~ increases at a decreasing rate, ~~suggesting that the~~ and faces a ~~greater~~ flatter gradient as the velocity increases toward 19.5m s^{-1}, suggesting that the opposing force increases proportionate to velocity. [2]

X

(ii) Determine the magnitude of the drag force F_D acting on the ball at time t = 3.5 s.

1 / 3

at $t = 3.5$s, $V =$ ~~15.25 ms^{-1}~~ $16.25\ ms^{-1}$

~~At~~ $m = 8.0kg$ $F_w = (8.0)(9.81)$

~~At~~ ~~8.98~~

using (3.4, 16.0),

$F_{net} = (8.0)(2.5)$

$\text{gradient} = \frac{16.25-16.0}{3.5-3.4}$

$= 2.5$ ~~x~~ $1.7\ ms^{-1}$

$F_D = F_w - F_{net}$

$= (8.0)(9.81-2.5)$

$= 58.48$

$= 58N$ (2 s.f.)

~~At~~ gradient $\frac{dv}{dt} = a$

$F_D =$ ..... 58 ..... ~~x~~ N [3]

(b) The drag force F_D is given by the equation

$$F_D = 6\pi r \eta v$$

where r is the radius of the ball,
η is the viscosity of air,
v is the velocity of the ball.

(i) Determine the SI base units of η.

3 / 3

$F_D = 6\pi r \eta v$

$kg\ m\ s^{-2} = 6\pi\ m^2\ s^{-1}\ \eta$

$6\pi\eta = kg\ m^{-1}\ s^{-1}$

base SI units is $kg\ m^{-1}\ s^{-1}$

base units = ... ~~kg m^{-1}~~ $kg\ m^{-1}\ s^{-1}$ ..... [3]

(ii) Explain how the maximum drag force will change if a ball of a greater mass, but the same radius, is released from rest from the same building.

0 / 1

The maximum drag force does not change as drag force at max is unrelated to mass. ~~It is dependent on~~

..... [1]

[Total: 9]

2 **(a)** State what is meant by *coherent* light waves.

0 / 1

Waves of ~~same~~ constant wavelength, phase, amplitude, intensity and are in phase with each other [1]

(b) Coherent light of wavelength 530 nm is incident normally on a double slit, as illustrated in Fig. 2.1.

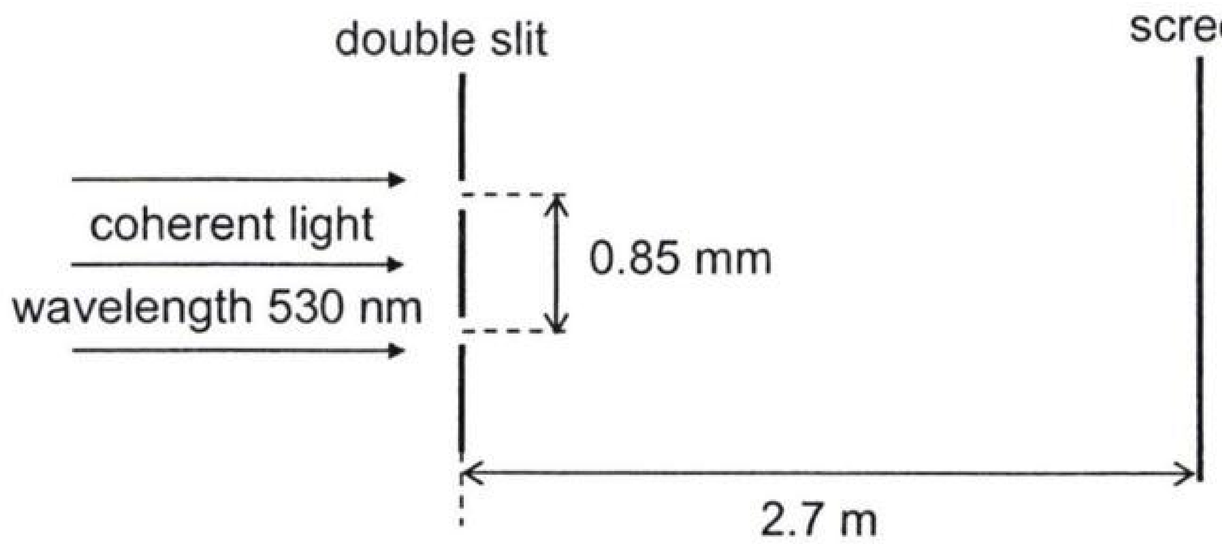


Fig. 2.1 (not to scale)

The separation of the two slits is 0.85 mm. A screen is placed parallel to the plane of the double slit at a distance of 2.7 m from the double slit.

Determine the separation of the fringes observed on the screen.

0 / 2

$\sin\theta = \frac{n\lambda}{a}$

separation = ... m [2]

(c) Describe the changes to the appearance of the fringes when each of the following changes is made separately.

(i) The width of both slits is reduced without altering their separation.

1 / 1

The bright fringes will be dimmer. [1]

(ii) The screen is rotated so that it is no longer parallel to the plane of the two slits.

0 / 2

..... [2]

[Total: 6]

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3 In an internal combustion (i.c.) engine, there are cylinders with movable pistons. Each cylinder contains a gas which can be treated as a monatomic ideal gas.

During the thermodynamic process of the i.c. engine called compression stroke, the gas in a cylinder is compressed rapidly from an initial volume of 6.0×10^{-4} m^3 to a final volume of 5.0×10^{-5} m^3. The initial temperature and pressure of the gas was 300 K and 1.0×10^5 Pa respectively.

(a) Calculate the amount, in mol, of the gas in the cylinder.

0 / 2

$pv = nrt$ SHOW substitution

$(1.0 \times 10^5)(6.0 \times 10^{-4}) = n ? (5.0 \times 10^{-5} \times 300)$

amount = ... mol [2]

(b) At the end of the compression stroke, the gas reaches a temperature of 550 K.

Determine the pressure of the gas.

0 / 2

pressure = ... Pa [2]

(c) Calculate the increase in internal energy of the gas due to this compression.

0 / 2

increase in internal energy = ... J [2]

(d) (i) State the first law of thermodynamics.

0 / 1

.....

.....

..... [1]

(ii) The efficiency of the i.c. engine depends on the temperature difference between the start and end of the compression stroke. The greater the temperature difference, the higher the efficiency of the engine.

Use the first law of thermodynamics to explain why the compression stroke is designed to take place rapidly for higher efficiency.

0 / 3

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

4 **(a)** An experiment is performed using two large vertical plates maintained at potential + 6.0 V and – 6.0 V and separated by distance *d* as shown in Fig. 4.1.

When a particle of mass 0.25 g and charge *Q* is released from rest between the plates, it moves in a straight line at an angle of 45° to the vertical.

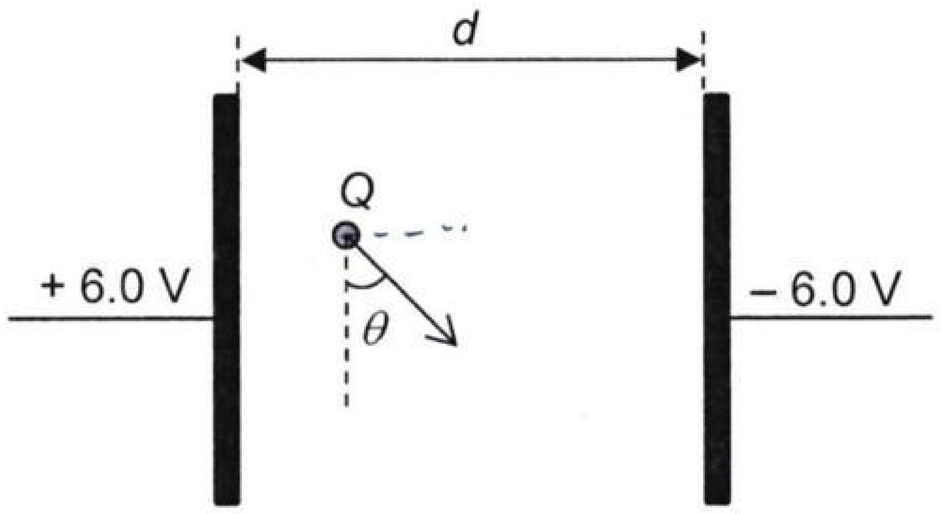


Fig. 4.1

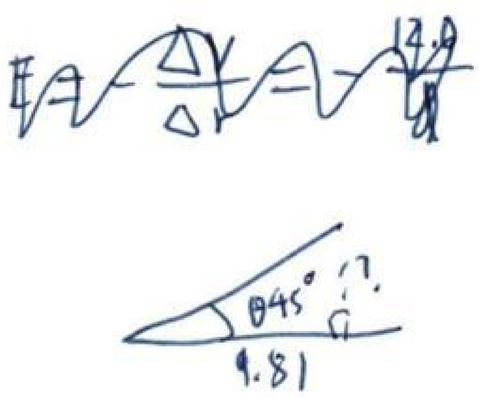
(i) State and explain the sign of the charge on the particle.

1 / 1

Q is positive because it is attracted to negative potential plate, and unlike poles attract. [1] ✓ (1)

(ii) Determine the magnitude of the electric force acting on the charged particle.

0 / 1



by vector resolution for weight & e-force:

electric force = $9.81 \tan 45^\circ$
$= 9.81$

Huh?

electric force = 9.81 N [1]

(b) The distance d between the plates is changed and the experiment is repeated with the particle being released from rest at the same initial position.

The graph of $\tan\theta$ against $\frac{1}{d}$ is shown in Fig. 4.2.

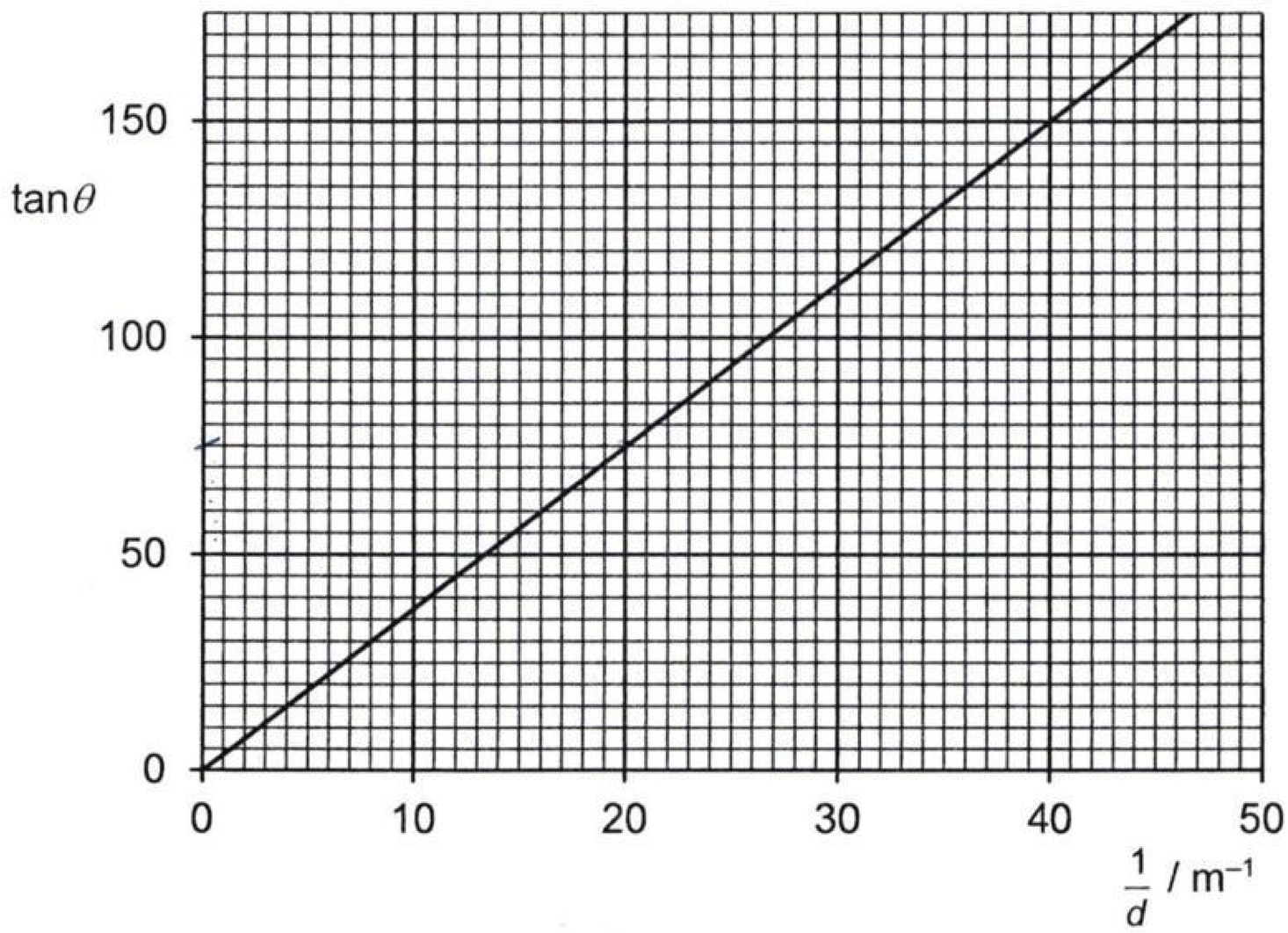


Fig. 4.2

Determine the magnitude of charge Q.

0 / 3

$E = -\frac{\Delta V}{\Delta r} = -\frac{12.0}{d} = (-12.0)\left(\frac{1}{d}\right)$

tan θ is coefficient of e-force ? Huh ?

e-force = QE

at $\frac{1}{d} = 20$, tan θ = 7.5

e- force = (9.81)(75) ?

E = (-12.0)(20)

$Q = \left|\frac{(9.81)(75)}{(-12.0)(20)}\right|$ Q is positive

= 3.0638

= 3.1 C (2 sf)

1 dp for 1/2 small sq

Q = ... 3.1 ..... C [3]

(c) The two large vertical plates in **(a)** form a parallel plate capacitor of capacitance 5.0 pF that is charged to 12.0 V.

This capacitor is then connected to another capacitor with capacitance of 10.0 pF that is initially charged to 6.0 V as shown in Fig. 4.3.

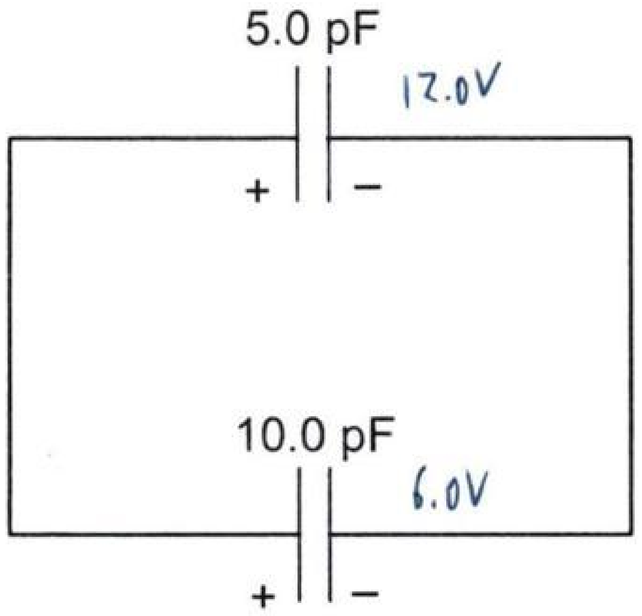


Fig. 4.3

Determine the final charge on each capacitor.

0 / 3

$C = \frac{Q}{V}$

$5.0\ pF = 5.0 \times 10^{-12}\ F$

$10.0\ pF = 10.0 \times 10^{-12}\ F$

Caps in series: $\frac{1}{C_T} = \frac{1}{C_5} + \frac{1}{C_{10}}$

$C_T = \left(\frac{1}{5\times10^{-12}} + \frac{1}{10\times10^{-12}}\right)^{-1}$

$= \frac{10}{3} \times 10^{-12}\ F$

$Q = CV$

$= \left(\frac{10}{3} \times 10^{-12}\right)(12.0)$

$= 4.0 \times 10^{-11}\ C$

charges:

$Q_5 = (5.0 \times 10^{-12})(12.0)$

$Q_{10} = (10.0 \times 10^{-12})(6.0)$

same charge $Q_5 = Q_{10}$

Final charges

$Q = (5.0 \times 10^{-12})(?)$

$Q_{5\,tot} = \frac{1}{2} Q_{5t}$

$= 4 \times 10^{-11}\ C$

$Q_{10\,tot} = \frac{2}{3}(5.0 \times 10^{-12})(12.0)(2)$

$= 8.0 \times 10^{-11}\ C$

final?

charge on the 5.0 pF capacitor = 4.0×10^{-11} C

charge on the 10.0 pF capacitor = 8.0×10^{-11} C [3]

[Total: 8]

5 A length of conducting wire is bent into the shape of an isosceles right-angled triangle, where the two equal sides are each 0.15 m in length as shown in Fig. 5.1.

The loop is placed in a uniform magnetic field of flux density 0.40 T directed perpendicularly into the plane of the loop. The loop moves at a constant velocity of 0.30 m s^{-1} in the plane of the loop to the right.

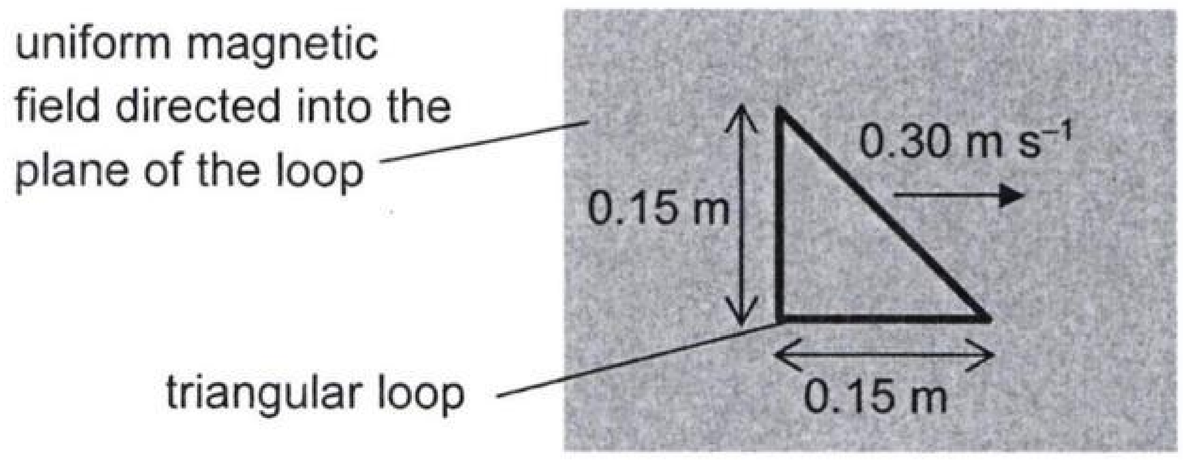


Fig. 5.1

(a) Explain why an e.m.f. is induced in the loop as it moves out of the magnetic field.

0 / 2

By Faraday's law of electromagnetic induction, there is induced e.m.f when there is a change in magnetic flux. When an area of wired loop moves, there is a change in the magnetic flux By the loop and this induces e.m.f.

vague

[2]

(b) The loop starts to exit the field at time $t = 0$ and it fully exits the field at time $t = T$.

On Fig. 5.2, sketch the variation of induced e.m.f. E with time t.

0 / 1

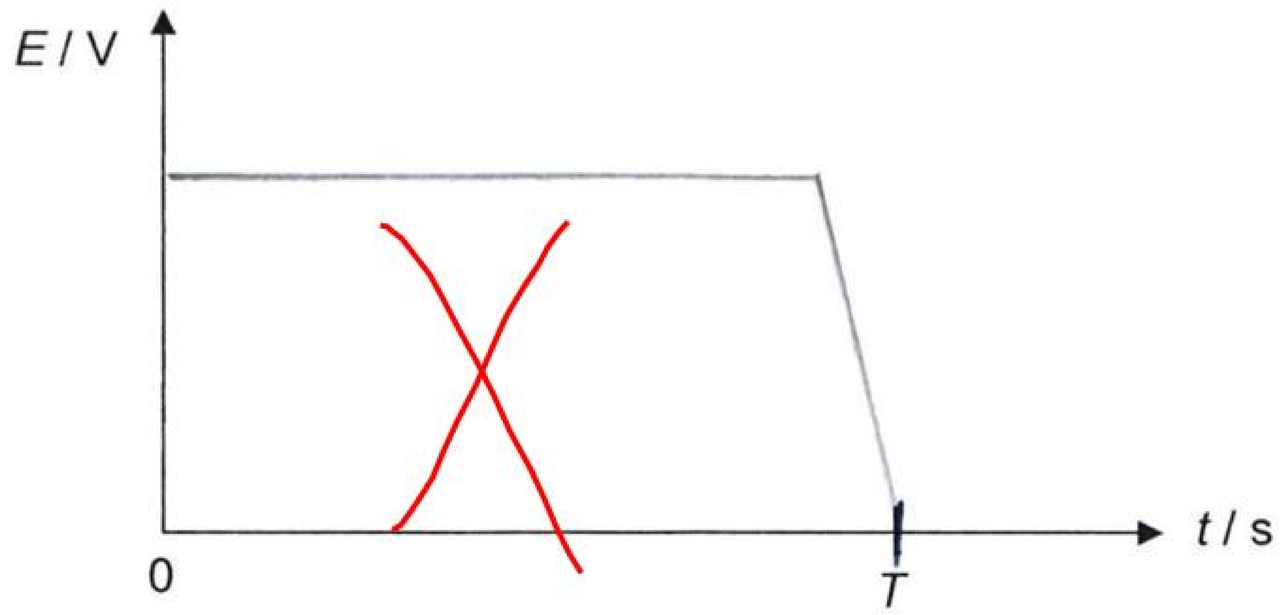


Fig. 5.2

[1]

(c) Determine the average e.m.f. induced in the loop.

0 / 2

e.m.f. = ... V [2]

[Total: 5]

6 **(a)** Describe and explain an observation from the photoelectric effect that provides evidence for the particulate nature of electromagnetic radiation.

0 / 2

.....

.....

.....

.....

.....

.....

..... [2]

(b) Explain how emission line spectra provide evidence for discrete electron energy levels in isolated atoms.

0 / 2

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.....

.....

.....

..... [2]

[Total: 4]

7 **(a)** State what is meant by the binding energy of a nucleus.

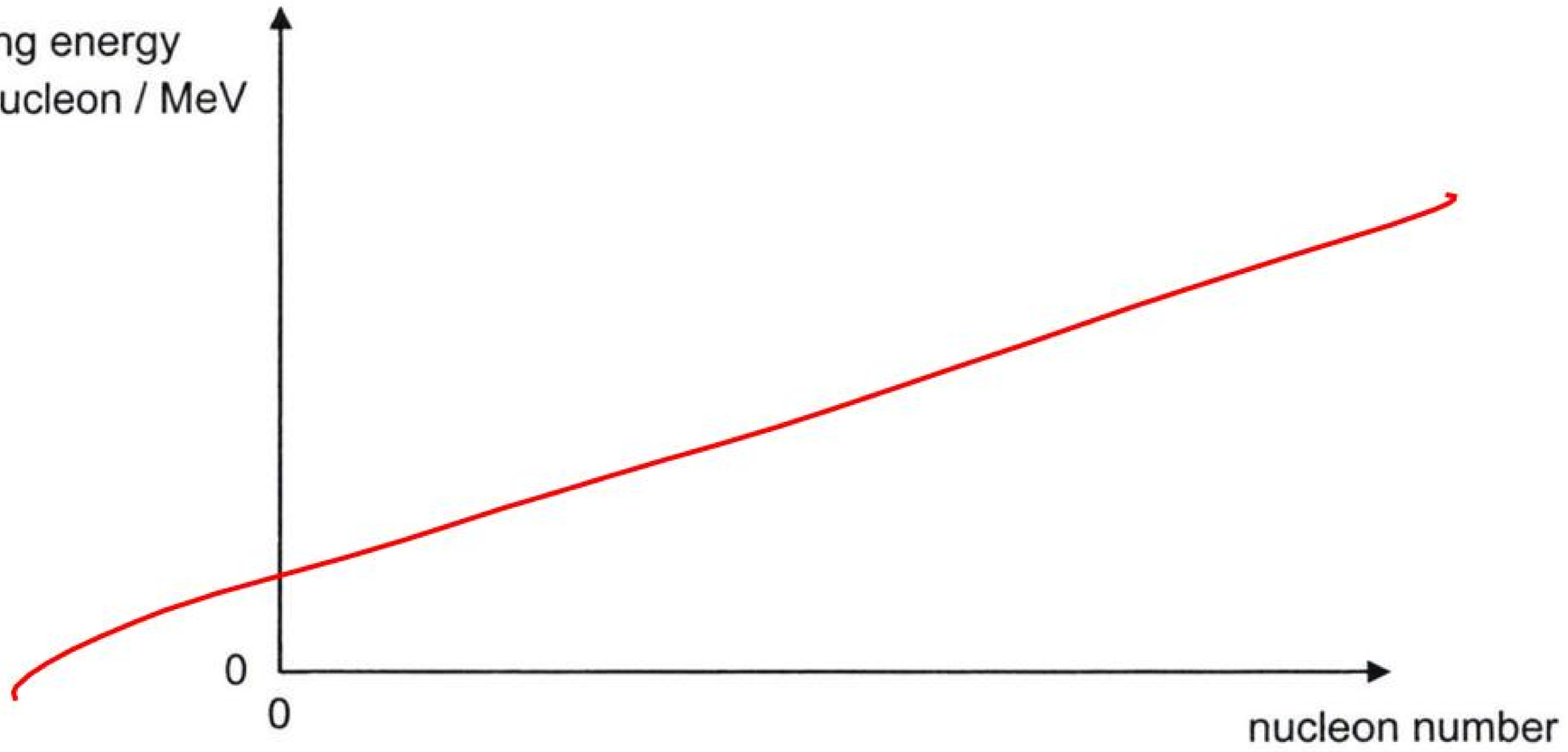
all? BOD

The amount of energy required to separate each nucleon from the nucleus to infinity. [1]

1 / 1

(b) The nuclear binding energy per nucleon varies with nucleon number.

On Fig. 7.1, draw a graph showing this variation for nuclei with nucleon number from 1 to 250. Label the axes with appropriate values.



0 / 3

Fig. 7.1 [3]

(c) Use your answer in **(b)** to explain why the fission of a nuclide having a high nucleon number is associated with a release of energy.

When high nucleon number nuclides go through fission, the resultant products have a lower mass. This gives a positive Δm X $m_f - m_i > 0$? , and since $E = \Delta m \times B.E.$, the positive change in mass produces a positive energy on the resultant side, releasing energy from the nuclear fission. [2]

0 / 2

(d) A fissile material is an isotope capable of sustaining a nuclear fission chain reaction when struck by neutrons. Thorium-232 ($^{232}_{90}Th$) is used in certain nuclear reactors. Thorium-232 absorbs a neutron and then undergoes radioactive decay to form fissile uranium-233 ($^{233}_{92}U$).

(i) State the number of α-particles and β-particles emitted in this radioactive decay.

0 / 1

number of α-particles = 4 ✓

number of β-particles = 4 ✓ [1]

(ii) Explain why it is not feasible for thorium-232 to absorb a helium-3 ($^{3}_{2}He$) nuclide to form fissile uranium-235 ($^{235}_{92}U$).

0 / 1

It may start to go through nuclear fission during uranium-235 formation.

[1]

(e) When a nucleus of uranium-233 absorbs a neutron in a nuclear reactor, it can undergo fission to form various products and release further neutrons. In one such fission, xenon-137 and strontium-94 are produced together with the release of three neutrons. The reaction is shown in the equation.

$$^{233}_{92}U + ^{1}_{0}n \rightarrow ^{137}_{54}Xe + ^{94}_{38}Sr + 3\,^{1}_{0}n$$

Data for the nuclei in the reaction are given in Fig. 7.2.

nucleus	mass / u
neutron $^{1}_{0}n$	1.008664
uranium $^{233}_{92}U$	233.039634
xenon $^{137}_{54}Xe$	136.911557
strontium $^{94}_{38}Sr$	93.915355

Fig. 7.2

Use data from Fig. 7.2 to determine, to three significant figures, the energy associated with the change in mass in this reaction.

1 / 3

$\{(233.039634 + 1.008614) - (136.911557 + 13.915355 + 1.008664 \times 3)\}$

$= 0.195394$ u

$\Delta m = 0.195394$ u

energy = ... J [3]

[Total: 11]

8 Read the passage below and answer the questions that follow.

In recent years, the global shift towards electric vehicles (EVs) has been driven by the need for sustainable transportation. EVs are often considered 'greener' and more efficient than conventional internal combustion engine (ICE) vehicles. In terms of energy consumption, a typical EV consumes approximately 18 kW h per 100 km, while a conventional petrol vehicle with an ICE consumes about 7.0 litres of fuel per 100 km. Petrol used in ICE vehicles has an energy density of approximately 34.2 MJ per litre.

An EV has a battery that supplies direct current, which an inverter converts into alternating current to drive its electric motor. In the motor, coils of wire are arranged around a rotating part called a rotor and placed in a magnetic field. When current flows through these coils, magnetic forces act on opposite sides of the coil in opposite directions, producing a torque to spin the motor. Fig. 8.1 shows one of the coils.

As the motor spins, a back electromotive force (e.m.f.) is induced in the coils which oppose the supply voltage and plays an important role in regulating the current drawn by the motor.

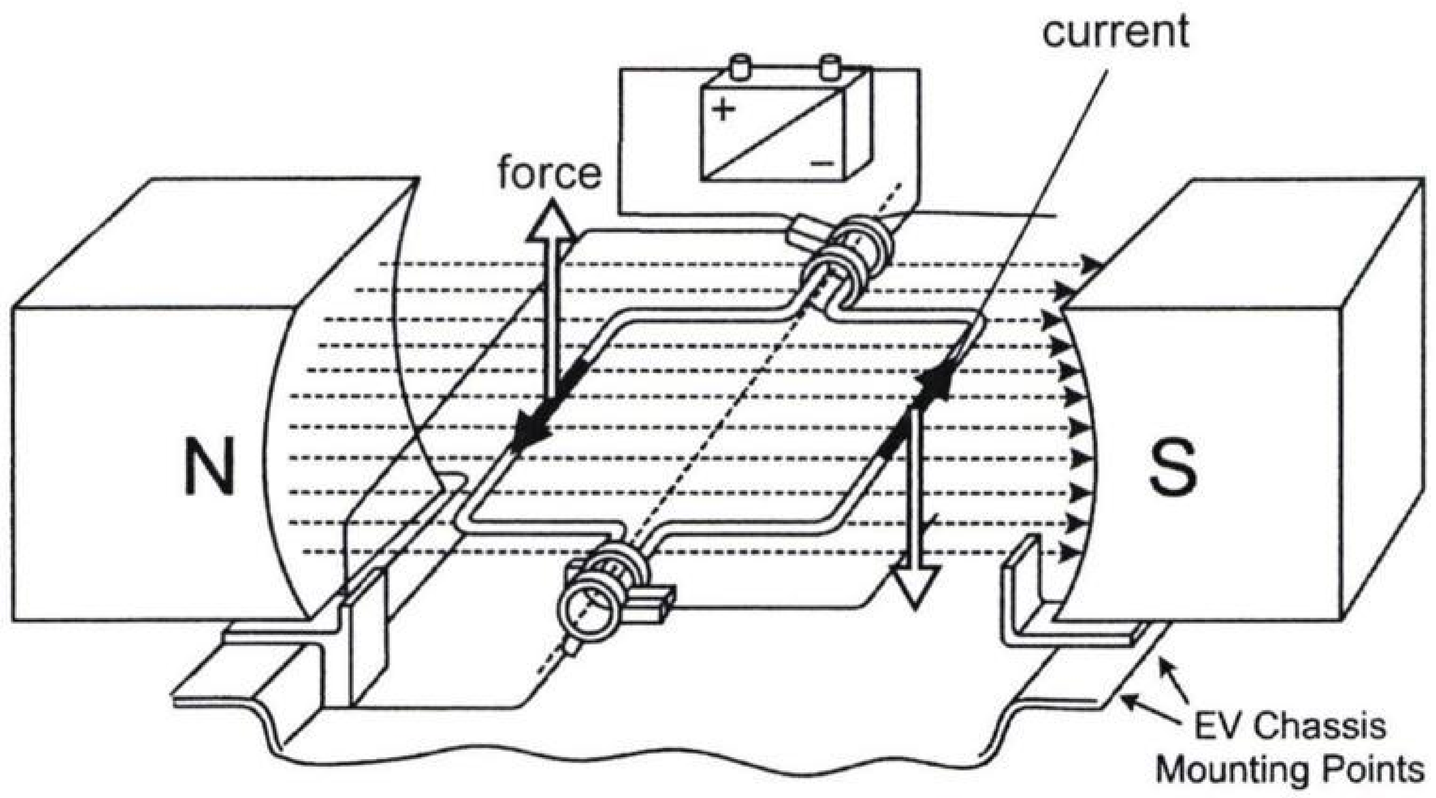


Fig. 8.1

Efficiency in an EV is further enhanced by regenerative braking. When the driver brakes, the motor's role is reversed and it acts as a generator, converting the vehicle's kinetic energy into electrical energy. As chemical batteries cannot absorb these rapid current surges without overheating, ultracapacitors are used as buffers.

Ultracapacitors are energy storage devices with very high capacitance and very low internal resistance. Since an individual ultracapacitor cell has a low voltage rating of 2.8 V, multiple cells are connected in series and parallel to form a high voltage ultracapacitor module needed by an EV.

Engineers tested a prototype ultracapacitor module with internal resistance, by charging it from a completely uncharged state with a constant current of 120 A from a power supply. Fig. 8.2 shows a simple circuit of the connection between the constant current power supply to the prototype ultracapacitor module.

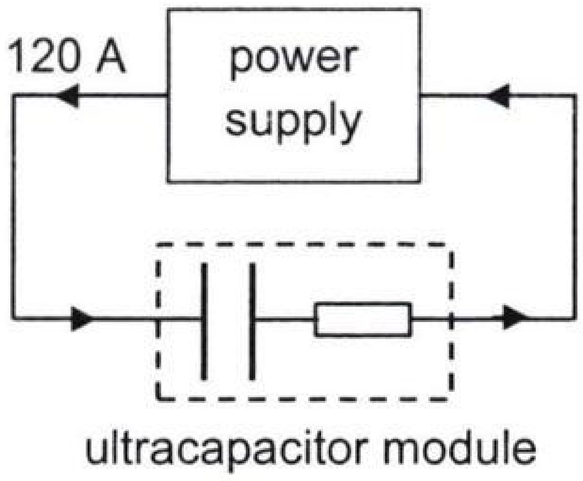


Fig. 8.2

The total voltage V_T across the ultracapacitor module was recorded over time t as shown in Fig. 8.3.

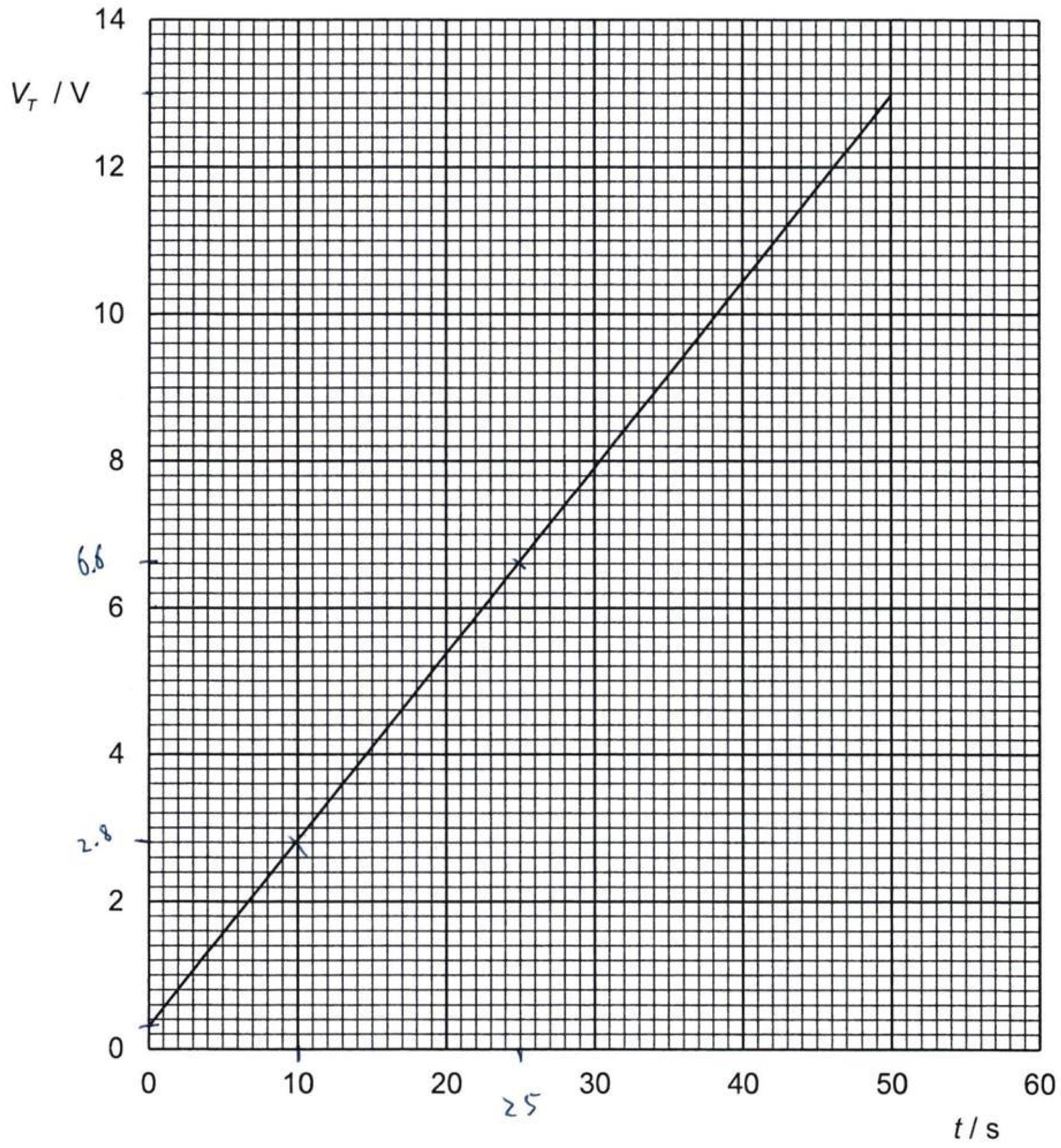


Fig. 8.3

(a) (i) For a petrol vehicle, calculate the average power required if the vehicle travels 100 km in one hour at constant speed.

1 / 2

work done = (7.0) (34.2 x 10^6)

time = 1 x 60 x 60 = 3600

power = $\frac{\Delta W}{\Delta t}$

= $\frac{(7.0)(34.2 \times 10^6)}{3600}$

= 665000 W

= 670 kW (2sf.)

power = ... 670 ..... kW [2]

(ii) The electricity for an EV is generated by a power plant operating at 50 % efficiency.

By comparing the energy per kilometer required by an EV and a petrol vehicle, explain why the EV is still considered 'greener'.

Support your explanation with appropriate calculations.

1 / 2

at 18 kWh x 2 = 36 kWh per 100km

while petrol vehicle is 670 kWh per 100km

eck

EV uses significantly less energy than petrol vehicle for same distance and same speed, so it is more energy efficient and does not waste as much as petrol vehicles.

..... [2]

(b) As the EV accelerates, the motor spins faster.

(i) Explain why the current in the motor decreases as the motor spins faster.

0 / 2

As the speed of the coil rotating in the magnetic field increases, the rate of change of magnetic flux in the coil increases as well, which induces a back e.m.f by Faraday's law of electromagnetic induction.

..... [2]

(ii) At a certain constant speed, the back e.m.f. is 300 V.

If the supply voltage is 360 V and the total coil resistance is 0.60 Ω , calculate the mechanical power developed by the motor.

0 / 2

voltage actually going through = 360 − 300 = 60V

$P = IV$

$= \frac{V^2}{R}$ ✗

$= \frac{(60)^2}{0.60}$

$= 600 = 0.60\text{kW}$

power = 0.60 ✗ kW [2]

(c) Using Fig. 8.3, determine for the ultracapacitor module,

(i) its internal resistance *r*,

1 / 1

~~at $t = 10$, $V = 2.8$~~ supply is $I = 120$

end charge is 13V at $t = 0$, $V = 0.3$

$V_0 = 13$

$2.8 = 13\left[1 - e^{-\frac{10}{I}}\right]$

$r = \frac{V}{I}$

$= \frac{0.3}{120}$ ✓

$= 0.0025$

r = 0.0025 ✓ Ω [1]

$\frac{2.8}{13} = e^{-\frac{10}{I}}$

$-10 = \ln\left(1 - \frac{2.8}{13}\right) = I$

(ii) the ideal voltage V_c across it at $t = 40$ s,

2 / 2

? Original voltage @ $t = 40$ is 10.4V.

in ideal cap, $r = 0$.

without resistance, P.D. drawn is 0.3V less

$V_c = 10.4 - 0.3$ ✓

$= 10.1$V

V_c = 10.1 ✓ V [2]

(iii) its capacitance *C*.

0 / 2

$V_0 = 13$

$2.8 = 13\left[1 - e^{-\frac{10}{I}}\right]$

$1 - \frac{2.8}{13} = e^{-\frac{10}{I}}$

$\frac{-10}{\ln\left(1 - \frac{2.8}{13}\right)} = I = R($

$\left(= \frac{-10}{(0.0025)\ln\left(1 - \frac{2.8}{13}\right)}\right.$

$= 1640.7$ ✗

$= 16000$ F (2 s.f.)

C = 16000 ✗ F [2]

(d) Each individual ultracapacitor cell has a capacitance of 3600 F and a voltage rating of 2.8 V.

(i) Determine the number of cells N that must be connected in series to support a voltage of 42 V across this series string.

1 / 1

voltage is split

$\frac{42}{2.8} = 15$

N = ... 15 ..... [1]

(ii) Calculate the effective capacitance of N cells in the series string.

1 / 1

$\left(\frac{1}{3600} \times 15\right)^{-1} = 240$

capacitance = ... 240 ..... F [1]

(iii) An ultracapacitor module consists of an arrangement of series strings each of N cells.

Compare your answers in **(c)(iii)** and in **(d)(ii)**,

State, with a reason, what can be deduced about the arrangement of the series strings of cells in the module.

1 / 1

It is a group of parallel series strings. Capacitor adds up in parallel, and putting 240F series strings in parallel adds to larger capacitance. [1]

(e) To test an emergency roadside charging system, engineers connected an uncharged ultracapacitor module across a constant 24 V DC power supply in series with the vehicle's climate control system, which acts as a constant resistive load of $0.50\ \Omega$.

Together, the ultracapacitor module and the climate control system form a simple RC charging circuit. The internal resistance of the ultracapacitor module is neglected during this test.

The variation of the voltage V across the ultracapacitor module with time t during this charging process is given by

$$V = V_0\left(1 - e^{-t/\tau}\right)$$

where is τ is the time constant of the RC circuit.

Use your value of C from **(c)(iii)** for the ultracapacitor module in the following calculations.

(i) Calculate the value of τ for the emergency charging circuit.

0 / 1

τ = ... s [1] 0

(ii) On the axes in Fig. 8.4, sketch a graph showing how the voltage V across the ultracapacitor module varies with time t from $t = 0$ to $t = 5\tau$.

Label the numerical value of V at $t = \tau$ and any other key numerical values on the vertical axis.

0 / 2

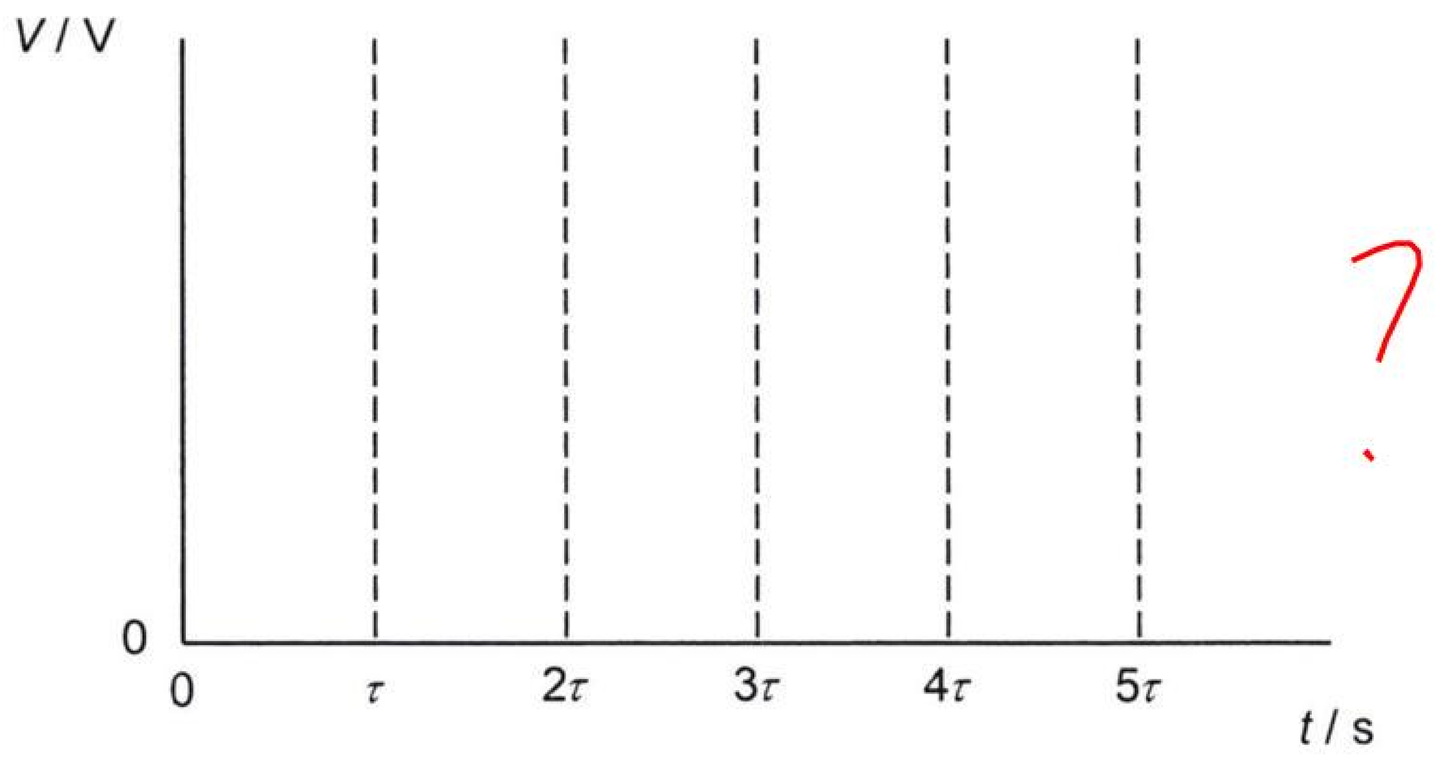


Fig. 8.4

[2] 0

(iii) The ultracapacitor module is allowed to charge for a long time until it is fully charged to 24 V.

1. Calculate the total electrical energy stored in the fully charged module.

0 / 1

energy = ... J [1] 0

2. Calculate the total energy delivered by the 24 V DC supply.

0 / 1

energy = ... J [1] 0

3. State a reason for the difference in the energy values calculated in **(e)(iii)1.** and **(e)(iii)2.**

0 / 1

..... [1] 0

[Total: 22]